Zaineb Messaoudi

Software Engineering Student · Full-Stack & Applied AI
Seeking a 6-Month PFE Internship January 2027

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SUMMARY

Final-year software engineering student at ESPRIT (Web & Internet Technologies) with internships across full-stack development and applied machine learning. Built and shipped production features in React, Node.js/NestJS, Django and Spring Boot for fintech, healthcare and edtech products, led the ML project of an HR intelligence platform (XGBoost, LightGBM, SHAP, $R^2 = 0.88$). Available for a 6-month final-year internship (PFE) from January, in software engineering, web development and applied AI/ML..

TECHNICAL SKILLS

Languages : Python, TypeScript, JavaScript, Java, PHP, C/C++, SQL, HTML, CSS
Frontend & Backend : React, Angular, NestJS, Node.js, Django, FastAPI, Spring Boot, Laravel, REST APIs, JWT, RBAC
AI / ML : LLM, RAG, Ollama, Hugging Face, NLP, Faster-Whisper, OpenCV, MediaPipe, scikit-learn, XGBoost, LightGBM, SHAP, Pandas
Data & real-time : MongoDB, PostgreSQL, MySQL, Redis, Meilisearch, WebSockets, Socket.IO, WebRTC
DevOps & QA : Docker, Git/GitHub, CI/CD, Jenkins, Kubernetes, Jira/Xray, SonarQube, Agile/Scrum
UI/UX & Multimédia : Figma, Canva, Adobe Photoshop, Adobe Illustrator,

PROFESSIONAL EXPERIENCE

Full-Stack Developer Intern · QCMed (remote) 06/2026 – 08/2026
- Built quiz and challenge modules (React, NestJS): detailed question attributes and granular tracking of medical residents' progress.
- Designed a Node.js REST API (MongoDB aggregations) feeding a real-time admin dashboard over WebSocket.
- Reworked challenge scoring to accept partially correct answers: less manual review and positive feedback from 60 % of testers.
- Expanded the residency exam bank to 294 sets across 18 specialties, with national ranking and instant correction.

Full-Stack / AI Developer Intern · Esprit Tech, Ariana 07/2025 – 08/2025
- Delivered HIKMALEARN, an accessibility platform for students with disabilities (React, Django REST), in 1.5 months, in a team of 6 (code reviews, 2-week agile sprints).
- Implemented 4 accessibility features: automatic video captioning, screen reader, keyboard navigation and AI-generated image descriptions.
- Shipped to production after passing a WCAG 2.1 audit, with 100 % unit-test coverage.

Web Developer Intern · Central Bank of Tunisia 02/2024 – 06/2024
- Built a secure back-office platform for cross-department event coordination (Spring Boot, Java, Thymeleaf, SQL) replacing an Excel process: collaboration cycle cut by 40 %.
- Implemented role-based access control, authentication and an audit trail; UX/UI prototyping in Figma.

Odoo Developer Intern · I-Way, Tunis 07/2022 – 08/2022
- Developed an Odoo module (Python, JavaScript) automating patient appointment booking: processing time per file reduced by 60 %; optimised PostgreSQL queries and caching (P95 < 500 ms).

PROJECTS

Oralis · AI-based oral presentation assessment (team of 5) 01/2026 – 05/2026
React, NestJS, FastAPI, MongoDB, WebRTC, Socket.IO, Docker
- Full-stack platform: student/teacher flows, rubric-based grading, JWT, RBAC, WebRTC recording; CI/CD, Jira/Xray, SonarQube.
- AI pipeline (Faster-Whisper, MediaPipe, OpenCV) measuring speech rate, pauses, filler words and visual delivery cues.

TalentLens · HR analytics platform powered by Machine Learning 01/2026 – 05/2026
Python, XGBoost, LightGBM, SHAP, FastAPI, React
- Led the ML module on 15,000 job postings: anomaly detection, salary prediction ($R^2 = 0.88$), segmentation, recommendation, SHAP explainability.
- Exposed models through FastAPI to React dashboards (recruiters, candidates, admins).

SolarShare · Equipment rental marketplace with AI 09/2026 – 10/2026
Laravel 12, MySQL, Redis, Meilisearch, Ollama, Docker
- Catalogue module (70 tests, 1,007 assertions) and production Docker CI/CD (PHP-FPM, Nginx, queue workers) with health checks and rollback.
- 7 AI features: natural-language semantic search, RAG chatbot, price and description generation, automatic categorisation, image validation.

Learnwise · AI-assisted learning platform 09/2026 – 10/2026
Django REST, Django Channels, React, WebSockets, MySQL, Ollama, Hugging Face
- Course management API: role-based permissions, versioning and rollback, prerequisite cycles, scheduled publishing, moderation, ZIP export.
- Real-time class presence and AI assistant with voice transcription and cited answers; learning analytics.

EDUCATION

Engineering Degree, TWIN (Web and Internet Technologies) · ESPRIT School of Engineering 2024 – 2027
Bachelor's in Computer Science & Multimedia · ISAMM 2020 – 2024

CERTIFICATIONS & LANGUAGES

Hedora Blockchain Certification (HCF, HBF, HCDA) · Hedora Association (10/2026) · **Associate Python Developer** · DataCamp (09/2026) · **Prompt Engineering and Generative AI** · Coursera (ODC) (06/2025) · **Microsoft Azure Fundamentals (AZ-900)** · Microsoft Learn (07/2024)
Arabic: native · **French**: fluent · **English**: B2 (Amideast certified)